

DFM ANALYSIS REPORT

Part Information

Part Name	nist-ftc-08-machined-part.step
Manufacturing Process	CNC Milling
Material	Aluminium

DFM Summary

DFM Score	0 / 100
Total Checks	157
Passed	139
Warnings	0
Critical	18
Information	0

Geometry Summary

Solids	1
Faces	270
Edges	705
Vertices	442
Volume	503592.175
Surface Area	191687.873
Length	311.595
Width	222.695
Height	48.26

Feature Summary

Total Detected Features: 158

DFM Violations

Feature	Rule	Status	Measured	Recommendation
CYLINDER_5	Hole Depth Ratio	Critical	25.7	Reduce the hole depth or increase the diameter.
CYLINDER_7	Hole Depth Ratio	Critical	39.99	Reduce the hole depth or increase the diameter.
CYLINDER_9	Hole Depth Ratio	Critical	25.7	Reduce the hole depth or increase the diameter.
CYLINDER_11	Hole Depth Ratio	Critical	39.99	Reduce the hole depth or increase the diameter.

The current depth-to-
The current depth-to-
The current depth-to-
The current depth-to-

Feature	Rule	Status	Measured	Recommendation	
CYLINDER_50	Hole Depth Ratio	Critical	51.4	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_54	Hole Depth Ratio	Critical	79.98	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_57	Hole Depth Ratio	Critical	51.4	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_61	Hole Depth Ratio	Critical	79.98	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_83	Hole Depth Ratio	Critical	100.0	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_86	Hole Depth Ratio	Critical	100.0	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_87	Hole Depth Ratio	Critical	50.11	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_88	Hole Depth Ratio	Critical	35.53	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_89	Hole Depth Ratio	Critical	50.11	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_90	Hole Depth Ratio	Critical	35.53	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_144	Hole Depth Ratio	Critical	39.13	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_146	Hole Depth Ratio	Critical	24.84	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_148	Hole Depth Ratio	Critical	39.13	Reduce the hole depth or increase the diameter.	The current depth-to-
CYLINDER_150	Hole Depth Ratio	Critical	24.84	Reduce the hole depth or increase the diameter.	The current depth-to-

Report Status: DFM violations detected